Experiment 25

Create a new GitHub repository for a personal project. Initialize the repository with a README.md file. Then, clone the repository to your local machine, make some changes to the README.md (e.g., add a project description), and push the changes back to GitHub.

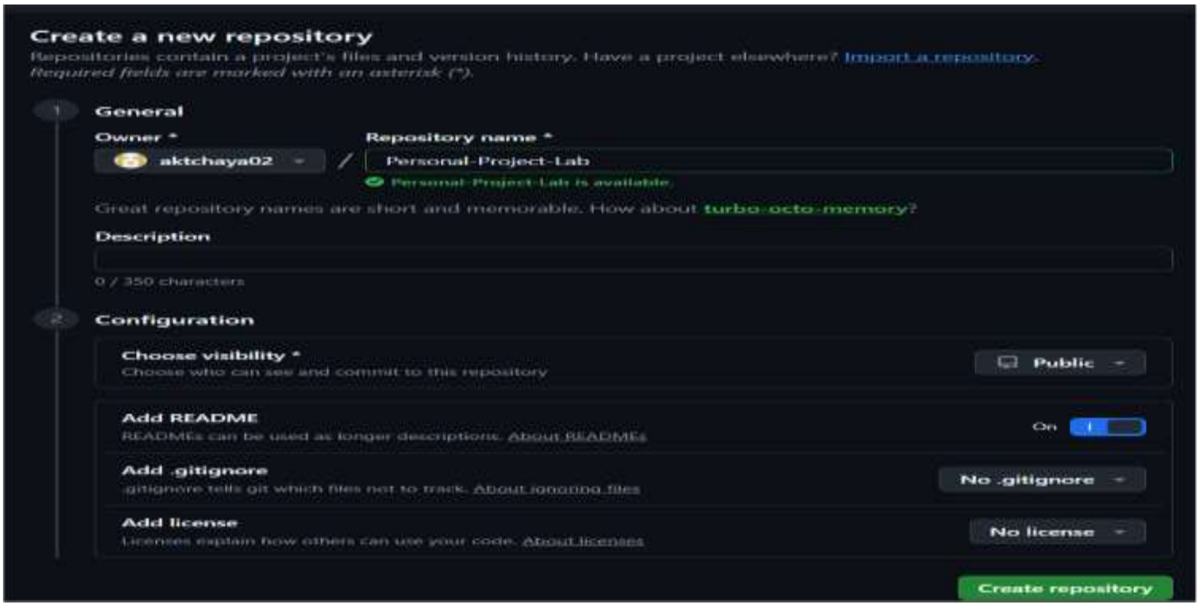
Tasks:

- **Create a GitHub repository.**
- **Clone the repository to your local machine.**
- **Edit the README.md file.**
- **Stage, commit, and push your changes to GitHub.**

Step 1: Create a New GitHub Repository

1. First, log in to GitHub.
2. Click the + icon in the top-right corner.
3. Select **New repository**.
4. Enter the repository name.

   Personal-Project-Lab
5. Select **Public**.
6. Select **Add a README file**.
7. Click **Create repository**.

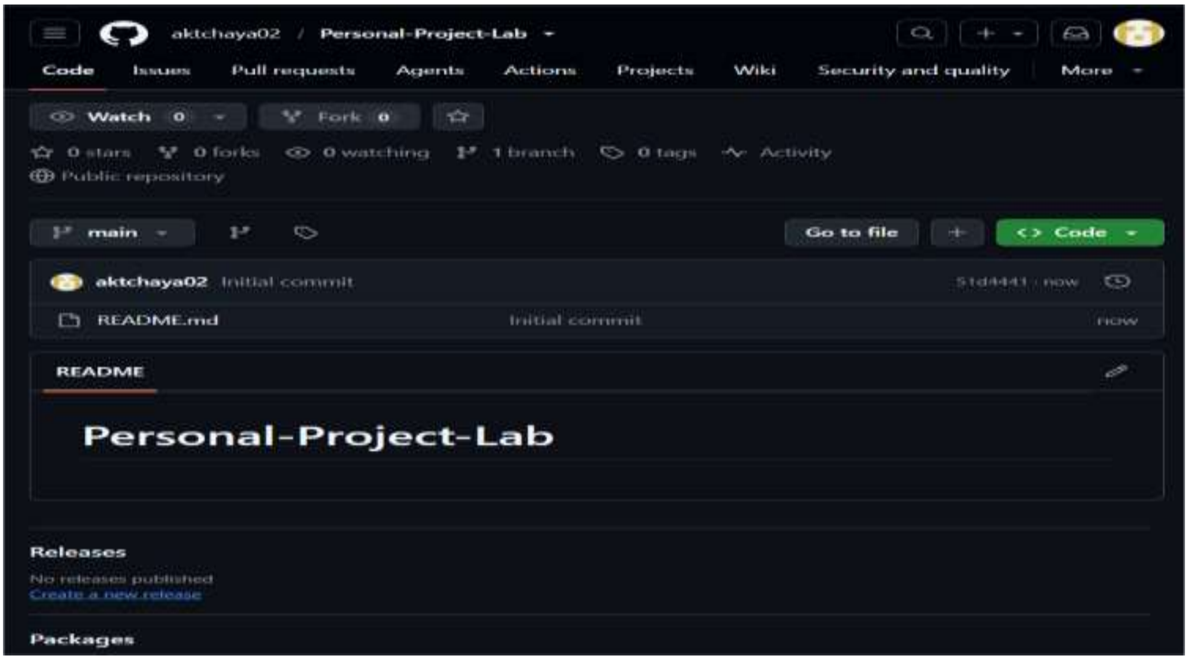


Step 2: Verify the Repository

1. After clicking **Create repository**, GitHub opens the repository homepage.
2. The repository should contain:

   README.md
3. The default branch should be:

   main
4. This confirms that the repository has been created successfully.



Step 3: Copy the Repository URL

1. Click the green **Code** button.
2. Select **HTTPS**.
3. Click the copy icon.
4. The URL will look similar to:

   https://github.com/YourUsername/Personal-Project-Lab.git

Step 4: Clone the Repository

1. Open **Command Prompt** on your computer.
2. Use the following command:

   git clone https://github.com/YourUsername/Personal-Project-Lab.git
3. Git downloads the repository from GitHub to your computer.

4. A successful clone will display messages similar to:

   Cloning into 'Personal-Project-Lab'...

   Receiving objects: 100%

   Resolving deltas: 100%

```
C:\Users\Hp>git clone https://github.com/aktchaya02/Personal-Project-Lab.git

Cloning into 'Personal-Project-Lab'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

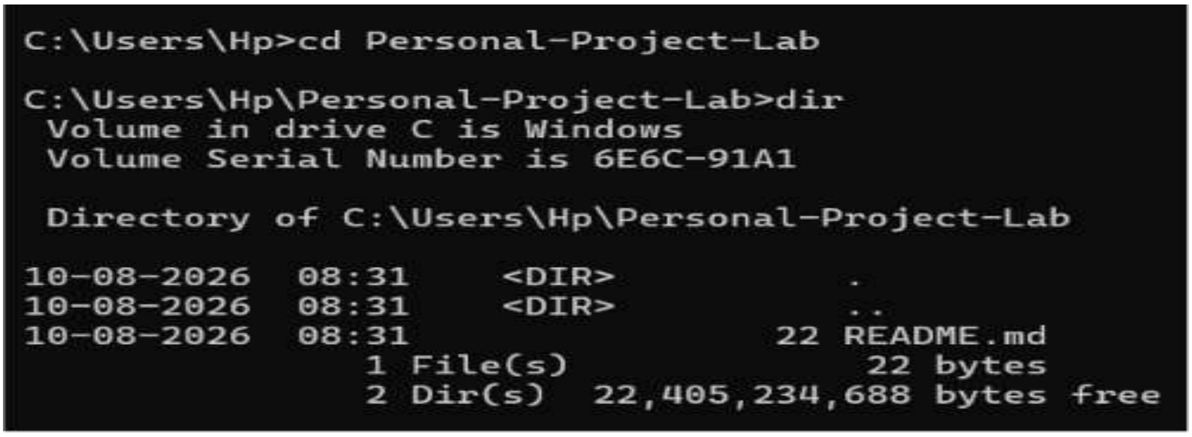
Step 5: Open the Local Repository

1. After cloning, move into the repository folder.
2. Run:

   cd Personal-Project-Lab
3. Then display the files:

   dir
4. You should see:

   README.md



```
C:\Users\Hp>cd Personal-Project-Lab

C:\Users\Hp\Personal-Project-Lab>dir
 Volume in drive C is Windows
 Volume Serial Number is 6E6C-91A1

 Directory of C:\Users\Hp\Personal-Project-Lab

10-08-2026  08:31    <DIR>          .
10-08-2026  08:31    <DIR>          ..
10-08-2026  08:31                22 README.md
               1 File(s)             22 bytes
               2 Dir(s)  22,405,234,688 bytes free
```

Step 6: Open the Repository in Visual Studio Code

1. From inside the repository folder, execute:

   code .
2. This opens the repository in Visual Studio Code.
3. In the VS Code Explorer, you should see:

   Personal-Project-Lab

   README.md

Step 7: Edit README.md

1. Open the README.md file in Visual Studio Code.
2. Add a project description.

   # Personal-Project-Lab

   ## Project Description

   This is my personal project repository created using Git and GitHub.

   The project demonstrates the basic Git and GitHub workflow,

   including repository creation, cloning, editing, committing,

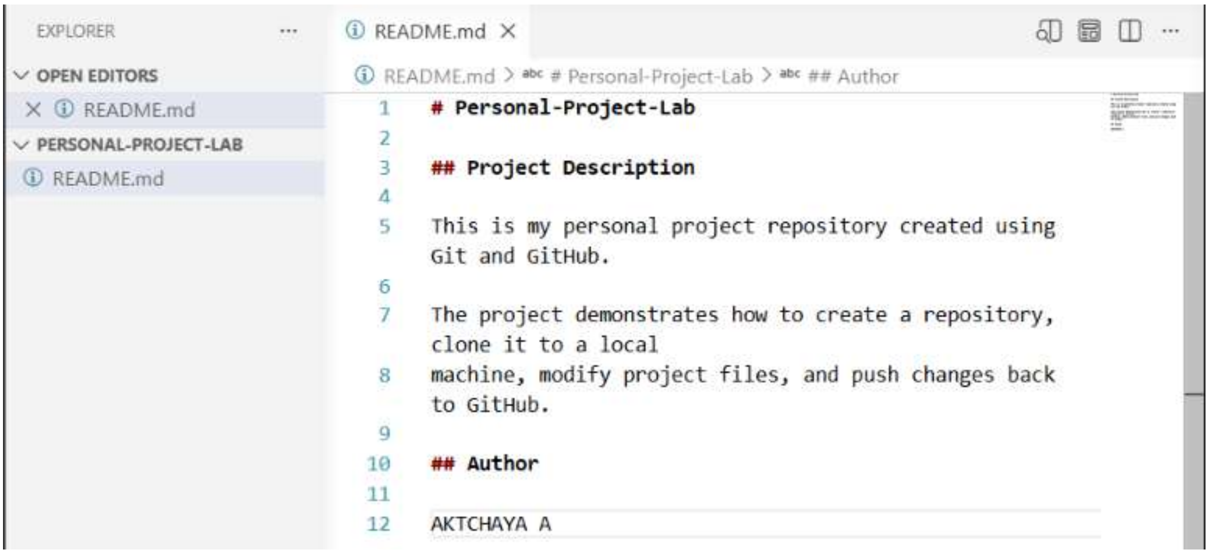
   and pushing changes.

   ## Author

   AKTCHAYA A

3. Save the file using:

   Ctrl + S



Step 8: Check the Repository Status

1. Return to Command Prompt.
2. Make sure you are inside the repository:

   C:\Users\Hp\Personal-Project-Lab

3. Run:

   git status

4. Git will show that README.md has been modified.
5. You may see:

   On branch main

Changes not staged for commit:

modified: README.md

```
C:\Users\Hp\Personal-Project-Lab>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

Step 9: Stage the Changes

1. Before committing, the modified file must be **staged**.
2. Run:

   git add README.md

3. Then check the status again:

   git status

4. You should now see:

   Changes to be committed:

   modified: README.md

```
C:\Users\Hp\Personal-Project-Lab>git add README.md

C:\Users\Hp\Personal-Project-Lab>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

Step 10: Commit the Changes

1. Now save the staged changes in Git's local history.
2. Run:

   git commit -m "Updated README with project description"

3. You should see output similar to:

   [main abc1234] Updated README with project description

   1 file changed

4. The commit message describes what was changed.

```
C:\Users\Hp\Personal-Project-Lab>git commit -m "Updated README with project
description"
[main 857f9e5] Updated README with project description
 1 file changed, 12 insertions(+), 1 deletion(-)
```

Step 11: Push the Changes to GitHub

1. To upload it to GitHub, run:

   git push origin main

2. Git sends the committed changes to the main branch on GitHub.
3. You should see output similar to:

   Enumerating objects...

   Counting objects: 100%

   Writing objects: 100%

   To https://github.com/YourUsername/Personal-Project-Lab.git

   main -> main

```
C:\Users\Hp\Personal-Project-Lab>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 461 bytes | 461.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/aktchaya02/Personal-Project-Lab.git
   51d4441..857f9e5  main -> main
```

Step 12: Verify the Changes on GitHub

1. Return to your GitHub repository and refresh the page.
2. Open README.md.

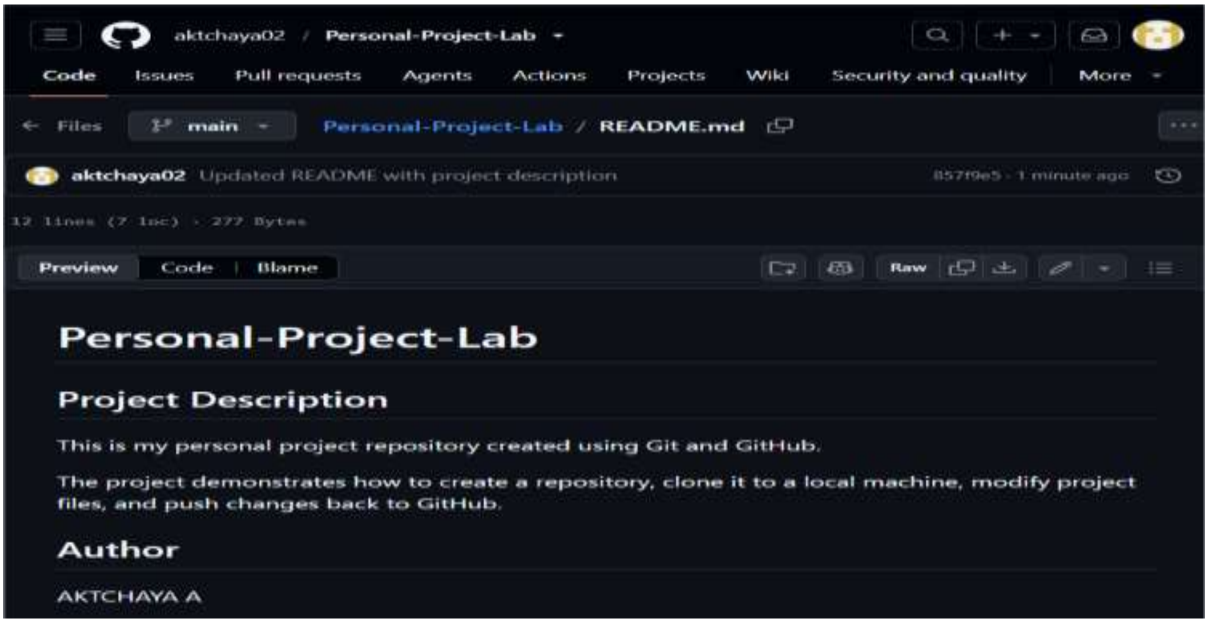
   The project description that you added locally should now be visible on GitHub.

3. For example:

   Personal-Project-Lab

   Project Description

   This is my personal project repository created using Git and GitHub.


aktchaya02 / Personal-Project-Lab
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
More
Files
main
Personal-Project-Lab / README.md
aktchaya02 Updated README with project description
857f9e5 · 1 minute ago
12 lines (7 loc) · 277 Bytes
Preview
Code
Blame
Raw
Personal-Project-Lab
Project Description
This is my personal project repository created using Git and GitHub.
The project demonstrates how to create a repository, clone it to a local machine, modify project files, and push changes back to GitHub.
Author
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